

Deep Learning Architectures for Automated Analysis and Classification of ECG Waveforms

(PhD Colloquium Viva)

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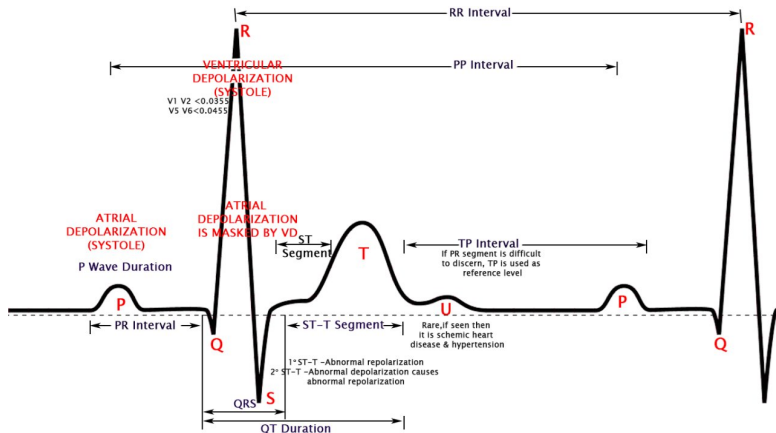
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Presentation Overview

- Introduction
- Literature Survey
- Motivation Headed Towards Research
- Research Objectives
 - Objective 1: Efficient Feature Extraction using Heterogeneous Activations (CHAF)
 - Objective 2: Hybrid CNN–Transformer Modeling for Global–Local Learning (ECG-ResViT)
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- Summary of Contributions
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Introduction - Electrocardiogram



Adapted from: P. S. Premanand and S. Narayanan, "Convolution-Based Heterogeneous Activation Facility for Effective Machine Learning of ECG Signals," *Computers, Materials & Continua*, vol. 77, no. 1, pp. 25–45, 2023. doi:10.32604/cmc.2023.042590

Literature Survey Overview

Automated ECG analysis can be broadly classified into the following categories:

- **Statistical ECG Analysis Methods**
 - Classical signal processing and statistical techniques
- **Machine Learning Approaches for ECG Classification**
 - Hand-crafted feature extraction + ML classifiers
- **Deep Learning Approaches for ECG Classification**
 - CNN, RNN, LSTM and advanced architectures
- **Hybrid and Advanced Architectures**
 - CNN–ML hybrids
 - CNN–Transformer models
- **Publicly Available ECG Datasets**
 - PhysioNet repository

Literature Survey

Author	Title	Method / Novelty	Metrics
Wehler et al., 2020	Frequency domain parameters applied onto ultra-short ECG recordings	HRV analysis using FFT and Lomb–Scargle Periodogram	High AUC for LF and HF
Liu et al., 2021	ECG quality assessment using statistical and S-transform features	Fusion of statistical features with CNN spectrogram features	Acc = 93.09%, F1 = 0.847
Singh et al., 2023	ECG signal feature extraction trends in methods and applications	Review of ECG feature extraction (time, frequency, time-frequency)	Review study
Barua et al., 2023	Hybrid handcrafted model for myocardial infarction classification	Multilevel pooling tree transformation with statistical + textural features	Acc = 99.94%
Zhou et al., 2023	ECG quantitative analysis using extremum energy decomposition	Extremum Energy Decomposition (EED) for ECG signal decomposition	Robust abnormality detection

Literature Survey (Contd...)

Author	Title	Method / Novelty	Metrics
Gowtham et al., 2024	Acquisition and analysis of cardiographic signal using transforms	IIR bandpass filtering, R-peak detection, spectral power extraction	Feature statistics reported
Alam et al., 2024	Statistical analysis of ECG pattern variations based on age and gender	ANOVA, Tukey HSD, and t-test on ECG features	Significant statistical variations
Darmawahyuni et al., 2024	ECG arrhythmia classification with feature optimization	Time-domain feature extraction with metaheuristic feature selection	Good Accuracy
Zou et al., 2022	Heartbeat classification using Random Forest with contextual feature	Segment-label feature extracted using CNN combined with Random Forest classifier	Acc = 0.96, F1 = 0.98
Sari et al., 2023	ECG signal classification using Random Forest for smart healthcare	R-R interval based ECG feature extraction with Random Forest classifier	Acc = 96%

Literature Survey (Contd...)

Author	Title	Method / Novelty	Metrics
Asmawati et al., 2024	Arrhythmia classification using Extreme Gradient Boosting	XGBoost based ECG classification model for arrhythmia detection	Acc $\approx$ 98–99%
Nawaz et al., 2024	ECG-based heart disease diagnosis using ML models	Comparative ML models: SVM, Random Forest, Logistic Regression, Decision Tree	Best Acc = 84%
Kalmady et al., 2024	ML algorithms for population-level cardiovascular diagnosis	ResNet-based deep learning with XGBoost ECG feature models	AUROC > 90% several CV
Wu et al., 2025	Deep learning and ECG for cardiovascular disease diagnosis	Systematic review of DL-based ECG diagnostic frameworks	Review study
Jaya Prakash et al., 2025	Advances in ML and DL for ECG beat classification	Systematic review covering CNN, RNN and hybrid ECG models	Review study
Gupta et al., 2024	AI models for abnormal cardiac rhythm classification	Review of lightweight and efficient AI architectures for ECG analysis	Review study

Literature Survey (Contd...)

Author	Title	Method / Novelty	Metrics
Bartusik-Aebisher et al., 2025	Artificial Intelligence and ECG for cardiac diagnostics	AI-based ECG monitoring using wearable and mobile healthcare devices	Review study
Hong et al., 2020	Deep learning methods for ECG data: opportunities and challenges	Systematic review of DL architectures for ECG analytics	Review study
Montenegro et al., 2022	Evaluation of ECG feature extraction using DL methods	Comparison of 1D-CNN and SVM for arrhythmia classification	CNN Acc $\approx$ 93%
Pósfay et al., 2023	ECG feature learning without backpropagation	Linear Law Transform (LLT-ECG) for automatic feature extraction	State-of-the-art results
Ahmed et al., 2024	Deep CNN-based ECG heartbeat classification	6-layer 1D-CNN architecture with batch normalization and dropout	Acc = 99.11%
Dhandapani et al., 2025	Hybrid deep learning framework for heart disease prediction	ANN-based DL architecture for automated ECG classification	Acc = 93.6%
Manimaran et al., 2025	Explainable DL techniques for ECG heart disease classification	XAI-based deep learning architectures with Grad-CAM, SHAP	Review study

Literature Survey (Contd...)

Author	Title	Method / Novelty	Metrics
Selvam et al., 2025	Hybrid CNN-VAE model for ECG classification	CNN feature extraction combined with Variational Autoencoder	Acc = 98.51%
Ghanim et al., 2025	Hybrid EfficientNetB7 and Vision Transformer framework	CNN morphological features with ViT global attention	Acc = 97.8%
Alyahya et al., 2026	Parallel Hybrid Model for ECG classification	Ensemble of EfficientNet, SequentialNet, and LeNet-5	Acc = 98.46%
Majid et al., 2026	SimCardioNet hybrid self-supervised learning framework	Multi-scale CNN with contrastive learning and attention	Acc = 97.5%, F1 0.972
Bhia et al., 2026	Comparative evaluation of DL models for CVD diagnosis	CNN, LSTM, ConvLSTM models with wavelet feature extraction	Acc = 99.98%, F1 98%
Jaya Prakash et al., 2025	Advances in ML and DL for ECG beat classification	Systematic review of CNN, RNN, and hybrid models	Review study

Literature Survey (Contd...)

Author	Title	Method / Novelty	Metrics
Kolhar et al., 2024	Hybrid ECG classification using AlexNet and dual-branch fusion network	Combination of AlexNet CNN and parallel dual-branch architecture	Acc $\approx$ 99%
Talukder et al., 2024	Hybrid ConvNeXt-X model for arrhythmia detection	ConvNeXt models combined with SMOTE-Tomek data balancing	Acc = 99.75%
Bai et al., 2024	CNN-BiGRU hybrid network for arrhythmia classification	CNN feature extraction with BiGRU and multi-head attention	Acc = 99.41%, F1 99.21%
Ye et al., 2025	CNN-BLSTM hybrid architecture for ECG arrhythmia detection	CNN for spatial features and BLSTM for temporal dependencies	Acc = 99.52%

Motivation Headed Towards Research

- Efficient extraction of discriminative ECG morphology
- Modeling of long-range temporal dependencies
- Need for composite and flexible ECG representations

- **Objective 1:** To design an efficient heterogeneous activation-based framework for robust ECG morphological feature extraction.
- **Objective 2:** To develop a hybrid CNN–Transformer architecture for modeling both local morphology and global temporal dependencies.
- **Objective 3:** To formulate a blending-based composite learning strategy for flexible and discriminative ECG representation.

Datasets Usage Across Objectives

- PTB Diagnostic ECG Database
(Objective 1: Feature Representation Learning)
- MIT-BIH Arrhythmia Database
(Objective 1: Feature Representation Learning)
- ARR-CHF-NSR Dataset (combined PhysioNet sources)
(Objectives 2 & 3: Classification and Evaluation)

Datasets Configuration for Objective 1

1. PTB Diagnostic ECG Database

- 549 records from 290 subjects (age: 17–87 years)
- 15 signals: 12-lead ECG + 3 Frank leads (V_x , V_y , V_z)
- Sampling rate: 1000 Hz, Resolution: 16-bit
- High-resolution clinical-grade dataset

2. MIT-BIH Arrhythmia Database

- 48 recordings (30 minutes each) from 47 subjects
- Two-channel ambulatory ECG recordings
- Sampling rate: 360 Hz, Resolution: 11-bit
- $\approx 110,000$ expert-annotated heartbeats

ECG Beat Extraction from Raw Signals

Motivation:

- Continuous ECG → Converted to beat-level signals
- Required for deep learning models

Methodology (Kachuee et al., 2018):

- 1 Segment ECG into 10s windows
- 2 Normalize amplitude
- 3 Detect R-peaks
- 4 Compute R-R interval
- 5 Extract heartbeat segment (1.2T)
- 6 Zero-pad to fixed length

IEEE ICHI Conference, 2018

3. ARR–CHF–NSR Combined Dataset (PhysioNet)

- Total 162 ECG recordings from 3 classes:
 - Arrhythmia (ARR): 96 recordings
 - Congestive Heart Failure (CHF): 30 recordings
 - Normal Sinus Rhythm (NSR): 36 recordings
- Sources:
 - MIT-BIH Arrhythmia Database
 - MIT-BIH Normal Sinus Rhythm Database
 - BIDMC Congestive Heart Failure Database
- Objective: Multi-class ECG classification (ARR vs CHF vs NSR)

Objective 1: Efficient Morphological Feature Extraction for ECG Classification

Problem Statement

- Deep CNNs provide strong ECG classification accuracy.
- However, they suffer from:
 - Large number of parameters
 - Dense-layer dominated computation
 - High inference latency
- Existing CNN–ML hybrid models:
 - Use homogeneous activation functions
 - Limit feature diversity

- Activation functions - fixed design choices.
- Most CNN-ML hybrids rely on a single activation (e.g., ReLU).
- The effect of **activation heterogeneity across convolution blocks** remains unexplored for ECG signals.

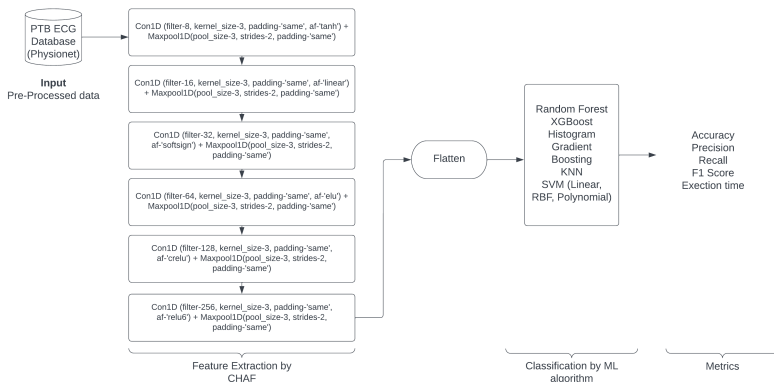
Question:

Can activation diversity improve ECG feature representation without increasing complexity?

CHAF: Convolution-Based Heterogeneous Activation Facility

- Uses CNN only as a feature extractor
- Assigns a different activation function to each convolution block
- Removes dense layers
- Employs lightweight ML classifiers for decision-making

CHAF ML Architecture



Adapted from: P. S. and S. Narayanan, "Convolution-Based Heterogeneous Activation Facility for Effective Machine Learning of ECG Signals," *Computers, Materials & Continua*, vol. 77, no. 1, pp. 25–45, 2023. doi:10.32604/cmc.2023.042590

Observation from Experiments:

Blocks	Accuracy (%)	Time (s)
4	98.53	25.00
5	99.07	32.68
6	99.18	47.12
7	98.80	84.84
8	98.69	390.10

Key Insight:

- Accuracy improves up to **six convolution blocks**
- Beyond six blocks: Accuracy and Computational cost
- Six blocks provide the optimal **bias–variance trade-off**

Activation Function Evaluation

Activation Function	Accuracy (%)	Time (s)
Swish	99.23	75.52
ELU	99.33	72.36
PReLU	99.16	77.77
SELU	99.04	52.84
CReLU	98.92	67.36
ReLU6	99.16	66.19
Leaky ReLU	99.27	68.32
Tanh	99.46	46.91
Sigmoid	72.14	45.48
GELU	99.34	54.41
Linear	99.49	44.59
Softsign	99.31	49.20

Why Heterogeneous Activation Functions?

Activation	Mathematical Form	ECG Interpretation
ReLU6	$f(x) = \min(\max(0, x), 6)$	Captures sharp and high-amplitude QRS complexes
Tanh	$f(x) = \tanh(x)$	Useful for modeling gradual variations in P and T waves
Softsign	$f(x) = \frac{x}{1+ x }$	Helping capture subtle waveform variations
ELU	$f(x) = \begin{cases} x & x > 0 \\ \alpha(e^x - 1) & x \leq 0 \end{cases}$	Capturing inverted or negative ECG deflections
CReLU	$f(x) = [\text{ReLU}(x), \text{ReLU}(-x)]$	Captures both positive and negative signal patterns
Linear	$f(x) = x$	Preserves raw signal morphology

Efficient Classification Strategy

Dense Layers in CNN:

- Add millions of parameters
- Dominate computational cost
- Increase inference latency

Key Observation:

- CNN convolution blocks already extract discriminative features
- Dense layers are not essential for ECG classification

Modification:

- ML classifiers IN! & Dense layers OUT! - Decision-making

Classifiers Used:

KNN

SVM

Random Forest

XGBoost

CNN vs CNN-ML Hybrid Performance

Model	Accuracy (%)	Time (s)
CNN + Dense	99.18	47.12
CNN-RF	99.10	3.05
CNN-XGBoost	99.12	2.56
CNN-KNN	99.06	0.006

Observation: ML classifiers achieve comparable accuracy with drastically lower inference time.

KNN Performance and Hyperparameter Optimization

Why KNN Performs Best

- CHAF produces compact and well-separated feature clusters
- KNN performs local, distance-based classification
- Minimal assumptions about data distribution
- No training overhead

KNN Hyperparameter Tuning

K Value	Accuracy (%)
3	99.48
5	99.55
7	99.50
9	99.42

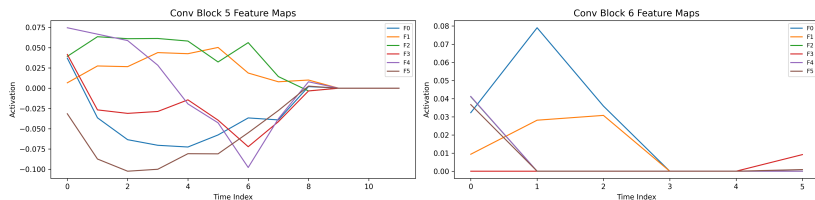
Observation: Optimal performance achieved at $K = 5$.

Feature Representation Perspective

- Each convolution block generates feature maps with distinct characteristics.
- Heterogeneous activations reshape feature distributions differently at each depth.
- Expected outcomes:
 - Reduced intra-class variance
 - Increased inter-class separability
 - Improved clustering in feature space

Observation: Reason why distance-based classifiers perform well.

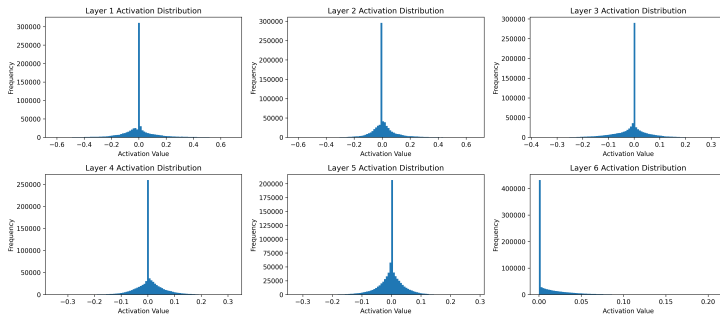
Hierarchical ECG Feature Representation Learned by CHAF



Observation

- Early convolution blocks capture signal transitions
- Intermediate layers capture waveform structures
- Deep layers learn abstract morphological patterns

Statistical Feature Distribution



Observation

- Activation variance reduces across convolution depth
- Conv1 variance: 0.0162
- Conv6 variance: 0.0004

Implication

- Progressive feature refinement
- Compact and stable representations

Mathematical Formulation

Convolution Operation

$$y_i^{(l)} = \sum_{k=1}^K w_k^{(l)} \cdot x_{i+k-1}^{(l-1)} + b^{(l)}$$

Activation Mapping

$$z_i^{(l)} = f_l \left(y_i^{(l)} \right)$$

where f_l denotes the activation function used in layer l .

CHAF Heterogeneous Activation Constraint

$$f_1 \neq f_2 \neq \dots \neq f_L$$

$$f_l \in \{\tanh, \text{ELU}, \text{CReLU}, \text{ReLU6}, \text{softsign}, \text{linear}\}$$

CNN with Dense Layers:

$$C = C_{\text{Conv}} + O(D \times M)$$

CHAF-ML:

$$C = C_{\text{Conv}} + C_{\text{ML}}$$

- Dense layers dominate CNN complexity
- Activation heterogeneity adds negligible cost

Novelty of CHAF

- To the best of our knowledge, attempts to systematically explore **layer-wise heterogeneous activation functions** within a CNN–ML hybrid framework.
- Demonstrates that **activation diversity** can improve feature representation without increasing model complexity.
- Combines:
 - CNN feature extraction
 - heterogeneous activation design
 - lightweight ML classifiers
- Achieves **high accuracy with real-time inference** for ECG classification.

Impact of Heterogeneous Activation Functions (CHAF)

Model	Accuracy (%)	Time (s)
CNN-KNN (ReLU)	99.06	0.006
CHAF-RF	99.09	10.08
CHAF-XGBoost	99.22	0.92
CHAF-KNN	99.55	0.008

Observation: Activation heterogeneity improves accuracy without increasing complexity.

Comparison with State-of-the-Art

Method	Dataset	Accuracy (%)	Time (s)
CNN	PTB	99.18	47.12
CNN-XGBoost	PTB	99.38	1.23
CHAF-KNN	PTB	99.55	0.008
CNN	MIT-BIH	99.06	295.26
CHAF-KNN	MIT-BIH	99.08	0.07

Observation: CHAF achieves superior accuracy with real-time inference.

Limitations of CHAF and Research Progression

Limitations

- Manual activation function selection
- CHAF focuses primarily on **local ECG morphology**
- Convolutional receptive fields limit modeling of **long-range temporal dependencies**

Observation

- Many cardiac abnormalities depend on **rhythm-level temporal relationships**
- Patterns often span **multiple cardiac cycles**

Motivation for Objective 2: Global Temporal Modeling using Hybrid CNN–Transformer Architectures

Objective 2: Hybrid CNN–Transformer Framework for Global ECG Temporal Modeling

Motivation: Beyond CHAF

- CHAF improves **local morphological feature diversity**
- However, ECG abnormalities often involve:
 - rhythm irregularities
 - inter-beat dependencies
 - long-term temporal evolution
- These patterns extend across multiple cardiac cycles

Need for explicit global temporal modeling

- **CNN-based models**

- strong local morphology extraction
- limited receptive field
- no explicit long-range dependency modeling

- **RNN / LSTM / GRU**

- sequential processing
- high computational complexity
- limited real-time feasibility

- **Transformer-only models**

- quadratic attention complexity
- heavy for long ECG signals

Key Question

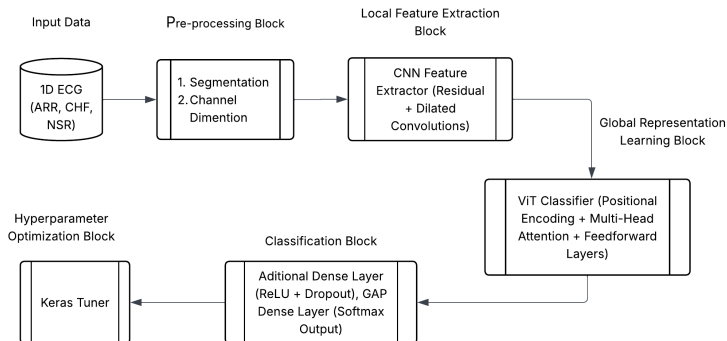
Can we design a computationally efficient hybrid architecture that captures both local ECG morphology and global temporal dependencies while preserving the native 1D signal structure?

- CNNs lack explicit global temporal modeling
- Transformer models are computationally expensive
- Existing hybrids lack efficiency and interpretability

Proposed Solution: ECG-ResViT

- Residual–dilated CNN
 - robust local morphology
 - multi-scale temporal features
- Vision Transformer
 - global temporal modeling
 - rhythm-level understanding
- Token-based design
 - reduced computational complexity

ECG-ResViT Architecture



Residual Learning for ECG Features

Residual connections enable stable deep learning.

$$y = F(x) + x$$

- preserves fundamental ECG morphology
- improves gradient flow
- enables deeper CNN feature extraction

Dilated Convolutions

Dilated convolution expands the temporal receptive field.

$$y(t) = \sum_{i=0}^{K-1} w(i)x(t - d \cdot i)$$

- dilation rates: $d = \{2, 4, 8, 16, 32\}$
- captures multi-scale temporal dependencies
- no increase in parameter count

Feature Tokenization

- CNN produces high-dimensional features
- Global Average Pooling compresses representation
- Linear projection converts features into token sequence

$$Z \in \mathbb{R}^{T \times D}$$

- T : temporal tokens
- D : embedding dimension

Vision Transformer for Global Modeling

Self-attention captures global temporal dependencies.

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- models rhythm-level ECG patterns
- captures long-range temporal dependencies
- parallel computation

Efficient Attention with Linformer

- Standard attention complexity: $O(T^2)$
- Linformer attention: $O(Tk)$ where $k \ll T$
- significantly reduces computation and memory
- scalable for long ECG sequences

Model	Parameters	Accuracy (%)
Base CNN-ViT	0.97M	96.30
ResDilated CNN-ViT	2.20M	97.78
ECG-ResViT	6.61M	99.13

Each architectural component contributes to performance improvement.

Comparison with CNN-Based ECG Classification Methods

Method	Approach	Accuracy
Wavelet + CNN + SVM	Scalogram-based CNN	98.81%
Transfer Learning CNN	CWT image-based CNN	98.7%
Hybrid AlexNet-SVM	Spectrogram + CNN + SVM	96.77%
Dual Branch CNN	CNN fusion architecture	99.0%
ECG-ResViT (Proposed)	CNN + Vision Transformer	99.13%

Key Insight

- CNN-based methods extract local ECG features effectively
- However, they lack global temporal dependency modeling
- ECG-ResViT integrates CNN and Transformer for joint local-global learning

Hybrid Model Analysis and Motivation

Key Advantages

- Joint local and global ECG modeling
- Preserves native 1D signal structure
- Computationally efficient hybrid architecture
- Interpretable via attention mechanisms
- Suitable for real-time clinical ECG analysis

Limitations

- Sequential processing of local and global features
- Lack of explicit feature blending

Motivation for Objective 3: Alpha-Blending Feature Fusion

Objective 3: Representation Blending for Robust ECG Classification

Motivation: Beyond Architectural Improvements

- Previous contributions improved ECG learning through
 - heterogeneous activation functions (CHAF)
 - hybrid CNN–Transformer modeling (ECG-ResViT)
 - 1D ECG representation
- However, ECG signals can be represented in multiple complementary forms
 - time–frequency spectral representation
 - structural morphological representation

Need for integrating complementary ECG representations

Problem Statement

- Most deep learning approaches rely on a single ECG representation
- Signal-based models capture waveform morphology
- Image-based models capture spectral characteristics
- Using only one representation limits feature diversity

Key Question

Can complementary ECG representations be integrated in a computationally efficient and interpretable framework?

- Existing fusion methods rely on
 - multi-branch architectures
 - attention-based fusion blocks
- These approaches increase architectural complexity

Comparison with Fusion-Based ECG Approaches

Method	Fusion Strategy	Accuracy
Bimodal CNN	Grayscale + Scalogram	95.74%
CWT Magnitude–Phase Fusion	Scalogram + Phasogram	98.5%
GAF-FusionNet	Multimodal Attention Fusion	99.6%
SIFT–CNN	Image Feature + CNN	99.78%
BlendNet (Proposed)	Alpha-Controlled Blending	100%

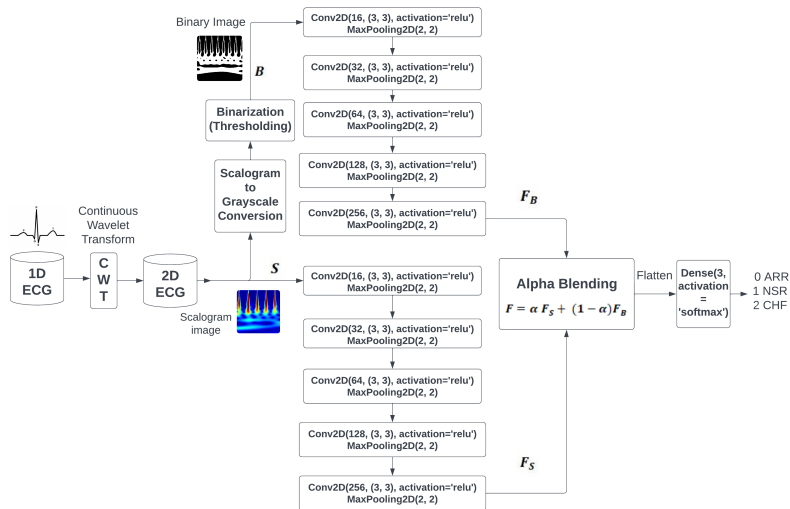
Key Insight

- Existing methods rely on complex multimodal fusion architectures
- BlendNet introduces a lightweight and interpretable feature blending mechanism

Proposed Solution: BlendNet

- BlendNet introduces a lightweight representation-blending framework
- Two complementary ECG representations are used
 - Scalogram (time–frequency features)
 - Binary structural map (morphological boundaries)
- Dual-stream CNN extracts features
- Alpha-controlled blending integrates both representations

BlendNet Architecture



Scalogram (CWT)

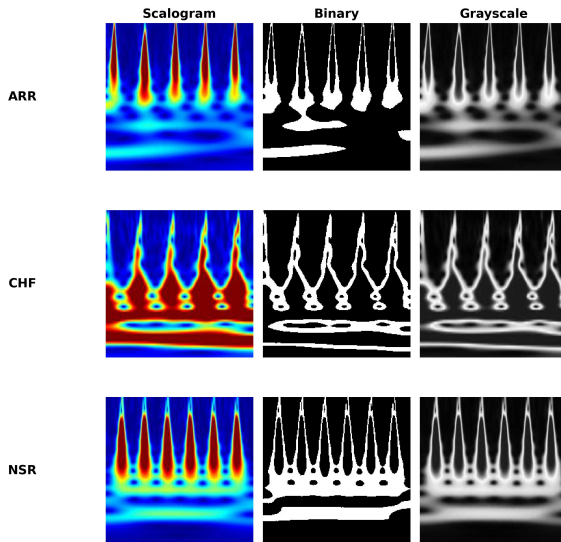
- Converts ECG into time–frequency representation
- Captures spectral and temporal dynamics
- Reveals frequency variations over time

Binary Structural Representation

- Highlights morphological boundaries
- Reduces intensity variations
- Emphasizes ECG structural patterns

Both representations capture complementary ECG characteristics

ECG Signal-to-Image Representations



Alpha-Controlled Feature Blending

Feature embeddings from two representations are fused.

$$F = \alpha F_s + (1 - \alpha) F_b$$

- F_s : scalogram-derived spectral features
- F_b : binary structural features
- α : blending coefficient controlling contribution

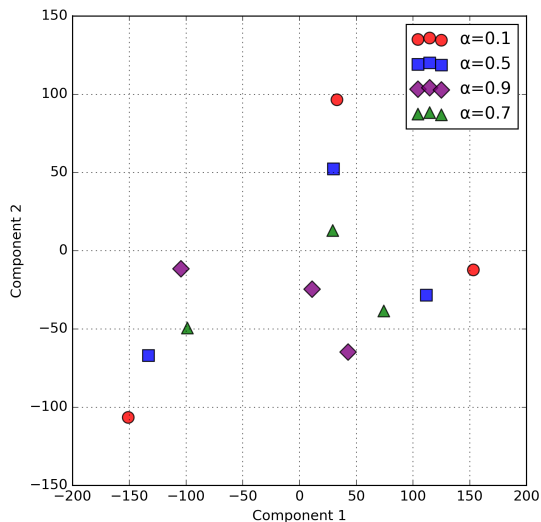
Baseline Performance: Single-Representation Models

Input Representation	Accuracy (%)	Execution Time (s)
Scalogram	96.29	272.17
Binary	91.11	263.24

Observation

- Scalogram representation achieves higher accuracy than binary-only representation
- Binary representation highlights structural ECG features but lacks spectral information
- Both representations capture **different ECG characteristics**
- This motivates the need for **feature-level blending**

Effect of Alpha Blending - Feature Embedding Visualization



Advantages of BlendNet

- Integrates complementary ECG representations
- Improves feature richness without deep fusion networks
- Lightweight and interpretable fusion mechanism
- Suitable for real-time ECG analysis

Summary of Contributions

Challenge	Learning robust representations from complex ECG signals
Contribution 1	CHAF : Introduces heterogeneous activation functions within CNN layers to enhance ECG morphological feature extraction
Contribution 2	ECG-ResViT : Hybrid CNN–Transformer architecture enabling joint local morphological learning and global temporal dependency modeling
Contribution 3	BlendNet : Alpha-controlled feature blending framework integrating spectral and structural ECG representations
Impact	Improved ECG representation learning for accurate and efficient automated cardiac diagnosis

- ❶ S. Premanand and S. Narayanan, "BlendNet: A Blending-Based Convolutional Neural Network for Effective Deep Learning of Electrocardiogram Signals," *Frontiers in Artificial Intelligence*, vol. 8, 2025. doi:10.3389/frai.2025.1625637.
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Thank You!